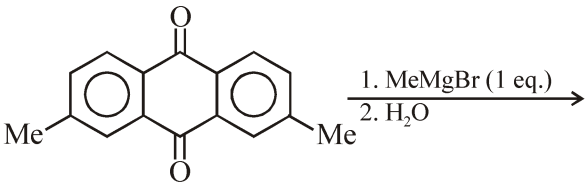
SECTION–I : (i) Only One option correct Type

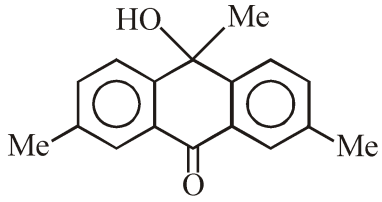
This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1.

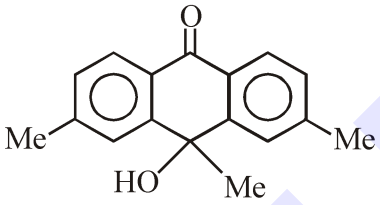


In above reaction out of the following which will be obtained in maximum yield.

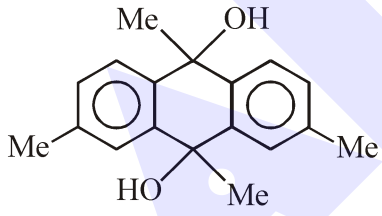
(A)



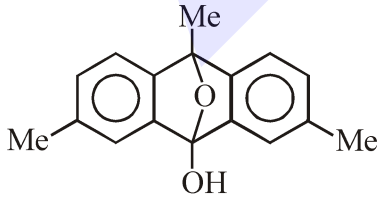
(B)



(C)



(D)



2. Select reaction which does not involve formation of free radical.

(A) Hundsdicker reaction

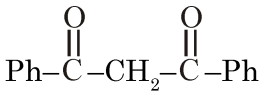
(B) Wurtz reaction

(C) Williamson's ether synthesis

(D) Kolbe's electrolysis synthesis

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3. Which of the following will not produce methane with MeMgBr ?

(A) 

(B) Me–C≡C–H

(C) H_2O

(D) 

4. Which of the following set of reagent(s) give cis-but-2-ene as a final major product starting with (+)-2-chlorobutane.

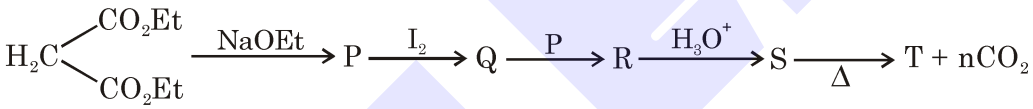
(A) Alcoholic KOH

(B) (i) CH_3COOAg , (ii) Δ

(C) (i) Alc. KOH , (ii) Br_2/CCl_4 , (iii) Alc. KOH ; $NaNH_2$, (iv) H_2/Pd-$BaSO_4$

(D) (i) Alc. KOH , (ii) Br_2/CCl_4 , (iii) Alc. KOH ; $NaNH_2$, (iv) Na/ Liq. NH_3

5. Find the value of **"n"** in following reaction sequence :



(A) 1 (B) 2

(C) 3 (D) 4

6. $CH_3–CH_3 \longrightarrow CH_4$

Identify reagent sequence for above conversion :

(A) (i) Cl_2 / hν ; (ii) Na / Et_2O

(B) (i) Cl_2 / hν ; (ii) Aqueous KOH ; (iii) PCC ; (iv) Red P / HI

(C) (i) Cl_2 / hν ; (ii) Aq. KOH ; (iii) $KMnO_4$, NaOH , Δ ; (iv) NaOH / CaO / Δ

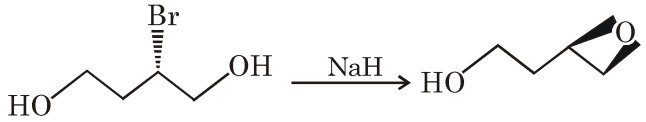
(D) (i) Br_2 / hν ; (ii) Alc. KOH / Δ ; (iii) O_3 ,Zn ; (iv) H_2, Pd

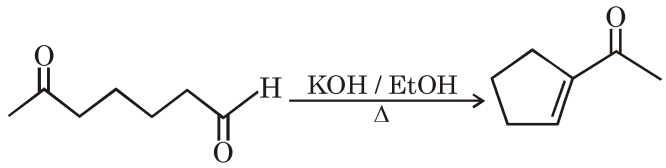
For More Material Join: @JEEAdvanced_2024

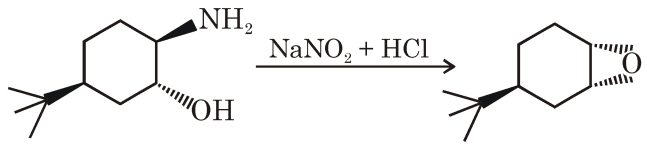
SECTION-I(ii) : One or More than one

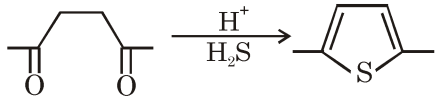
This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

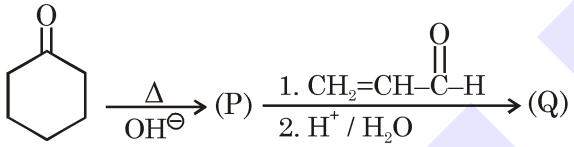
7. Identify major product formed are correctly matched in following reactions ?

(A) 

(B) 

(C) 

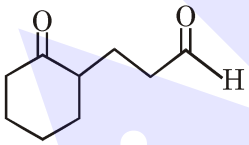
(D) 

8. 

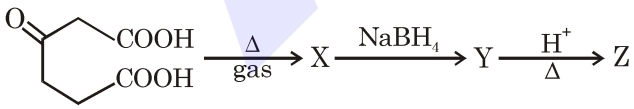
Correct statements for above reaction sequence :

(A) (P) is 2° alcohol

(B) (P) to (Q) involve 1,4 addition

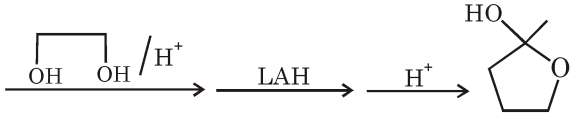
(C) (Q) is 

(D) (P) to (Q) involve 1,2 addition

9. 

Identify correct statement for above reaction sequence ?

(A) 'Z' product is sweet smelling substance

(B) Reaction of 'X' 

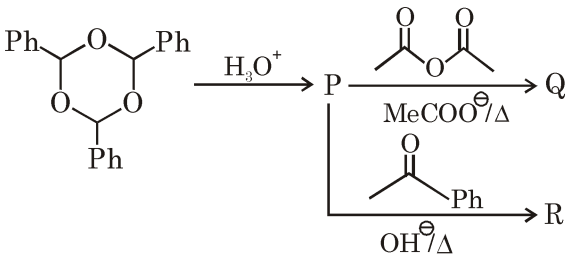
hemi ketal product

(C) 'Y' product librate CO_2 gas with aqueous $NaHCO_3$ and also soluble in aqueous NaOH

(D) 'Z' product is resolvable

For More Material Join: @JEEAdvanced_2024

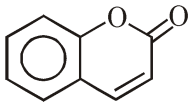
10.

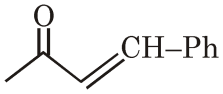


Identify correct statement for above sequence ?

(A)  is 'P' product

(B) Ph–CH=CH–COOH is 'Q' product

(C)  is 'Q' product

(D)  is 'R' product

SECTION–I(iii) : Paragraph Type

This section contains **2 paragraphs** each describing theory, experiment, data etc. **2 or 2 questions** relate to two paragraphs . Each question of a paragraph has **one OR more than one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

PARAGRAPH

Paragraph for Q.No. 11 to 12

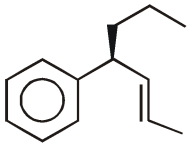
A optically active alkene (P) on reaction with H_2, Pt produces optically inactive (Q). (P) on reaction with 1% $KMnO_4$ followed by HIO_4 produces (R) & (S). When (R) & (S) are reacted in 1 : 2 respectively with NaOH, Δ, it produces (T) & HCOONa.

11. Which of the following can't possible as (P) ?

(A) 

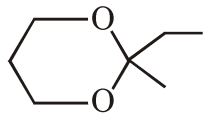
(B) 

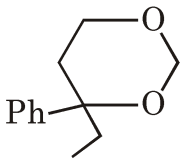
(C) 

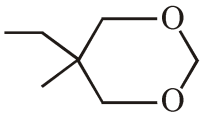
(D) 

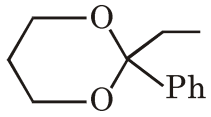
For More Material Join: @JEEAdvanced_2024

12. (T) on reaction with HCHO in acidic medium produces :

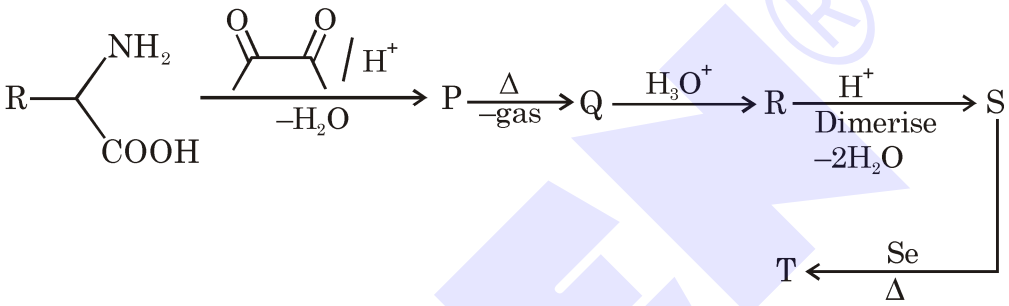
(A) 

(B) 

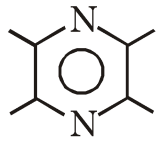
(C) 

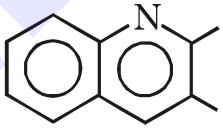
(D) 

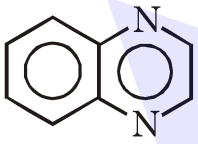
Paragraph for Q.No. 13 to 14

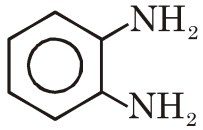


13. Identify 'T' product is above sequence ?

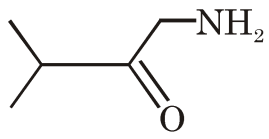
(A) 

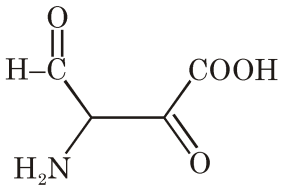
(B) 

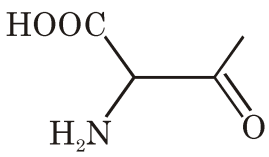
(C) 

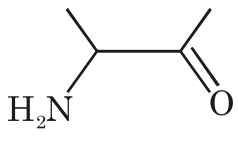
(D) 

14. Identify 'R' product in above sequence ?

(A) 

(B) 

(C) 

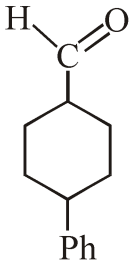
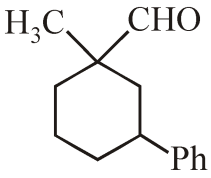
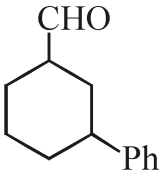
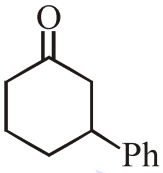
(D) 

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Matching List Type

This Section contains **1 multiple choice questions. Each question has matching lists.** The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

15.

List-I (Compound)	**List-II (Statements)**
(P) 	(1) It can show tautomerism, optical isomerism but not geometrical isomerism
(Q) 	(2) It can show tautomerism, geometrical isomerism but not optical isomerism
(R) 	(3) It can show geometrical isomerism , optical isomerism but not tautomerism
(S) 	(4) It can show tautomerism, geometrical isomerism & also optical isomerism

Code :

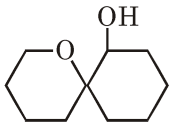
	P	**Q**	**R**	**S**
(A)	2	1	3	4
(B)	2	3	4	1
(C)	3	2	4	1
(D)	2	3	1	4

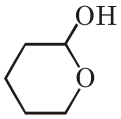
For More Material Join: @JEEAdvanced_2024

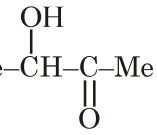
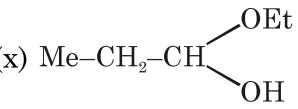
SECTION-III : (Integer Value Correct Type)

This section contains **04 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

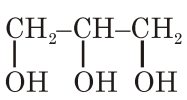
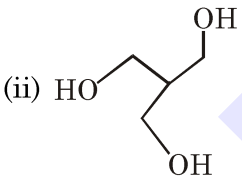
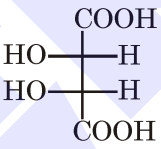
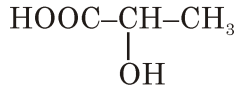
1. How many of the following will give silver mirror with tollen's reagent ?

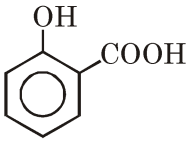
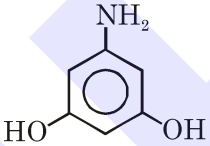
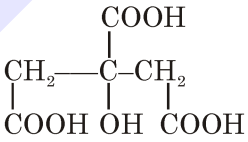
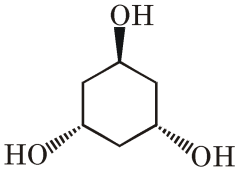
(i) Ph–CH=O (ii) HCHO (iii)  (iv) Me–C≡CH

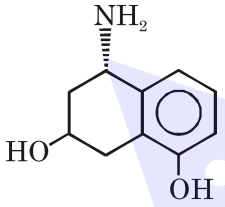
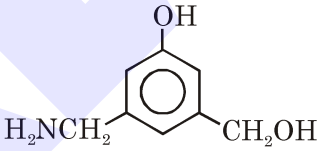
(v) HCOOH (vi) MeCOOH (vii)  (viii) HC≡CH

(ix)  (x) 

2. Total number of compounds whose molecular mass is incresed by 126 gm on reaction with excess CH_3–COCl / Pyridine :

(i)  (ii)  (iii)  (iv) 

(v)  (vi)  (vii)  (viii) 

(ix)  (x) 

3. A tetrapeptide on hydrolysis produced glycine (gly), Leucine (Leu.), Alanine (Ala) , Valine (Val.). Count number of possible tetrapeptides (structural isomers only) which can produce above four amino acids with the following both conditions.

(i) Glycine is not present at N-terminal (ii) Alanine in present at C-terminal

4. A segment (X) of cellulose obtained on partial hydrolysis has molecular mass 1476 gm. On complete acidic hydrolysis, mass of the product obtained is 1620 gm. Find out the number of glycosidic linkage(s) present in segment (X) :

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